# JUAN CAMILO MELO LÓPEZ

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## PROFESSIONAL PROFILE

Telematics Engineering student pursuing a career in Cloud/DevOps Engineering, with over a year of hands-on, project-based experience designing and automating end-to-end infrastructure: CI/CD pipelines, GitOps workflows, Infrastructure as Code, container orchestration, and observability across AWS and GCP. Backed by a broader foundation in networking, systems engineering, software development, and cybersecurity. Honor Roll student every semester to date, including three Honor Scholarship awards for top academic performance, reflecting a pragmatic, adaptable, and responsible approach to solving real infrastructure problems.

## EDUCATION

**Universidad Icesi** **Cali, Colombia**
*B.Eng. in Telematics Engineering (GPA: 4.7/5.0)* *Expected Feb. 2028*
Relevant coursework: Infrastructure I–III, Platforms I–II, Infrastructure Automation, Cloud Computing, Operating Systems, Software Engineering, Systems & Communications, IT Operations

**Academic Recognition:** Honor Scholarship (Beca de Honor) recipient for periods 2023-2, 2024-2, and 2025-2

## WORK EXPERIENCE

**Universidad Icesi** **Cali, Colombia**
*Networking Laboratory Assistant | Part-time* *Feb. 2026 – Present*

- Maintain and monitor the internal network infrastructure of the university laboratory, ensuring optimal performance and proper utilization of specialized equipment.
- Provide hands-on technical support during laboratory sessions, assisting faculty and students with device configuration, connectivity troubleshooting, and hardware maintenance.
- Build practical networking expertise by actively diagnosing and resolving connectivity issues and managing the configuration and specialized care of network devices.

*Academic Teaching Assistant: Algorithms & Programming | Part-time · Hybrid* *Feb. 2024 – Nov. 2024*

- Supported 30+ students per semester in designing and implementing algorithmic solutions in Java, reinforcing object-oriented design principles, data structure selection, and Git/GitHub version-control workflows.
- Guided students from problem specifications through algorithm design to working implementations, strengthening their ability to translate technical requirements into structured, maintainable code.
- Appointed by faculty based on top academic standing (GPA 4.7/5.0) and demonstrated technical proficiency in data structures and object-oriented programming.

## RESEARCH EXPERIENCE

**Research Group Celsia** **Feb. 2024 – Feb. 2025**
*Satellite Networks Researcher*

- Conducted research on the evolution and economic landscape of modern satellite network technologies, with a focus on Low Earth Orbit (LEO) constellations such as Starlink.

## PROJECTS

**Docket: GitOps Delivery Platform on AWS EKS** **Aug. 2026 – Sep. 2026**
Terraform · AWS EKS · ArgoCD · Kustomize · GitHub Actions (OIDC) · SonarQube · Trivy · Playwright · External Secrets · IRSA · CloudWatch · SNS · Lambda · Redis
*Team Coursework Project: Automatización de Infraestructura* · Repo ↗

- Delivered as a team five polyglot microservices (Go, Spring Boot, Node.js, Python, Vue) through ten connected pipelines, from commit to dev, staging, and production on a multi-AZ EKS cluster built from versioned Terraform modules.
- Implemented gated GitOps promotion with ArgoCD and Kustomize: SonarQube and Trivy HIGH/CRITICAL gates on every service, an integration suite before staging, a recorded approval before production, and rollback as a reverted PR.
- Secured and monitored it: CI reaches AWS via OIDC with no long-lived keys, secrets flow from SSM through External Secrets and IRSA, and CloudWatch alarms reach Slack via SNS and Lambda; 25 ADRs record every design decision.

**Platform Pipeline: Self-Hosted CI/CD, GitOps & Observability Platform** **Jul. 2026**
Go · Jenkins (JCasC) · Terraform · Docker · kind · ArgoCD · Helm · SonarQube · Gitleaks · Trivy · Syft (SBOM) · Prometheus · Grafana · Alertmanager · Gitea
*Personal Project* · Repo ↗

- Built a complete, locally runnable DevOps platform around a two-service Go application, with Jenkins fully configured as code (plugins.txt, JCasC, Job DSL) to run build, unit tests, a SonarQube quality gate, Gitleaks secret scanning, image build, Syft SBOM generation, and a Trivy vulnerability gate before pushing to a local registry.
- Provisioned the entire environment with Terraform using local state: Docker network, local registry, kind cluster, namespaces, and the ArgoCD and kube-prometheus-stack Helm releases, requiring zero cloud accounts or credentials.
- Implemented GitOps continuous delivery with ArgoCD reconciling a Kustomize-based deploy folder from a local Gitea instance; Jenkins releases by committing an image-tag bump, and a post-deploy health gate automatically reverts the release commit if the new version never becomes healthy.
- Set up full observability with kube-prometheus-stack, including a provisioned Grafana dashboard, ServiceMonitors, and PrometheusRule alerts with optional Alertmanager-to-Slack notifications; validated the loop with scripted traffic, pod-failure, and failed-release demos.

**E-Commerce Platform: Full Stack + AWS Infrastructure as Code** **Apr. 2026 – May 2026**

Java 21 · Spring Boot · Spring Security · JWT · React 18 · Tailwind CSS · PostgreSQL · AWS CloudFormation · EC2 ASG · RDS · ALB · CloudFront · S3 · CloudWatch · SNS · CloudTrail · MercadoPago

*Coursework Project: Infraestructura III* · *Repo ↗*

- Architected and deployed a full-stack e-commerce platform: a Java 21 + Spring Boot REST API with JWT/BCrypt authentication and role-based access control, a React 18 + Tailwind SPA served from S3 via CloudFront, and PostgreSQL on RDS, all provisioned through 10 CloudFormation stacks.
- Configured auto-scaling EC2 fleet (ASG min 1 / max 3) behind an ALB with CPU target-tracking policy; provisioned separate CloudFront distributions for frontend, backend proxy, and media assets with S3 OAC.
- Integrated MercadoPago Checkout Bricks payment gateway with HTTPS webhook callbacks; set up CloudWatch alarms, SNS email notifications, and CloudTrail audit logging for full operational observability.
- Automated the full deployment lifecycle with Bash scripts: sequential CloudFormation stack deployment, backend JAR upload to S3, frontend build + sync, and pre-demo environment health verification.

**IcesiScore: Serverless Real-Time Sports Platform** **Mar. 2026 – May 2026**

Python · AWS Lambda · API Gateway v2 (WebSocket) · Amazon Cognito · RDS PostgreSQL · DynamoDB · SNS · Terraform · Flutter · Clean Architecture + BLoC

*Coursework Project: Aplicaciones Móviles* · *Repo ↗*

- Built serverless backend on AWS: Python Lambda functions, Cognito JWT authentication, API Gateway REST + WebSocket endpoints, RDS PostgreSQL for relational data, and DynamoDB for WebSocket connection registry.
- Designed event-driven live-score broadcast pipeline: POST event → Lambda → SNS topic → broadcaster Lambda → DynamoDB scan → WebSocket postToConnection; managed stale connections via GoneException cleanup.
- Developed Flutter mobile frontend with Clean Architecture (domain / data / UI layers), GetIt dependency injection, and BLoC state management; provisioned all AWS infrastructure with Terraform.

**Kubernetes Cluster: Automated Provisioning with Ansible** **Mar. 2026**

Ansible · Kubernetes 1.32 (kubeadm) · Flannel CNI · containerd · BIND9 · dhcpd · Rocky Linux 9 · KVM/QEMU · GitHub Actions CI

*Coursework Project: Infraestructura III* · *Repo ↗*

- Automated end-to-end provisioning of a production-grade Kubernetes cluster (control plane + worker) on KVM/QEMU using idempotent Ansible playbooks with BIND9 DNS and DHCP bastion; achieved 0 configuration drift on repeated runs.
- Configured Flannel VXLAN overlay network, firewalld rules, and kube-proxy iptables for cross-node pod communication; validated NodePort service exposure and CoreDNS internal service discovery.
- Implemented GitHub Actions CI pipeline for automated Ansible playbook linting and syntax validation on every commit.

**ISP-Grade GPON Network: Full Infrastructure Stack** **Oct. 2025 – Nov. 2025**

MikroTik RouterOS · Cisco SG350X · Huawei OLT/ONT · Ubuntu Server 24.04 · BIND9 · Kea DHCP · Postfix · Dovecot · Caddy · Prometheus · Grafana · LibreNMS · LibreQoS

*Coursework Project: Infraestructura II & Plataformas I* · *Repo ↗*

- Designed a multi-VLAN dual-stack (IPv4/IPv6) ISP network integrating physical hardware (MikroTik, Cisco, Huawei GPON) with a full virtualized service stack on Ubuntu Server: BIND9 DNS, Kea DHCP, Postfix/Dovecot email, Caddy reverse proxy with load balancing, and LibreQoS per-client bandwidth management.
- Deployed Prometheus + Grafana + LibreNMS unified observability stack; configured SNMP monitoring and real-time alerting dashboards across all physical and virtual infrastructure.

## CERTIFICATIONS

**AWS Academy Cloud Operations** **Jul. 2026**
Amazon Web Services (AWS): Monitoring, automation, deployment operations

**AWS Academy Cloud Foundations** **May 2026**
Amazon Web Services (AWS)

**Google Cloud Computing Foundations Certificate** **Apr. 2026**
Google: Compute Engine, VPC networking, Cloud IAM, GKE fundamentals

**Google Cloud Skill Badges (4)** **2026**
Build a Secure GCP Network · Set Up App Dev Environment · Implement Load Balancing · Prepare Data for ML APIs

**Cisco DevNet Associate** **Nov. 2025**
Cisco: Network programmability, REST APIs, automation, CI/CD practices, Ansible, Python scripting

**Google Cybersecurity Certificate Series** **2024 – 2025**
Cybersecurity Foundations · Security Risk Management · Network Security · Linux & SQL

## TECHNICAL SKILLS

**Cloud Platforms:** AWS (EC2, ASG, EKS, RDS, S3, CloudFront, CloudFormation, CloudWatch, SNS, Lambda, API Gateway, Cognito, DynamoDB, CloudTrail), GCP (Compute Engine, VPC, Load Balancing, Cloud Run)

**DevOps & Automation:** GitHub Actions, Jenkins, ArgoCD (GitOps), Terraform, CloudFormation (IaC), Ansible, Docker, Kubernetes, Helm, Kustomize, SonarQube, Trivy, Bash scripting, CI/CD pipeline design, KVM/QEMU

**Backend Development:** Java, Spring Boot, Spring Security, JWT, REST APIs, Python, Go, PostgreSQL, JPA/Hibernate

**Frontend Development:** React, Vite, Tailwind CSS, JavaScript / TypeScript, Flutter

**Observability:** Prometheus, Grafana, Alertmanager, LibreNMS, AWS CloudWatch, SNMP monitoring

**Networking:** MikroTik RouterOS, Cisco IOS, VLANs, IPv4/IPv6 dual-stack, BIND9 DNS, Kea DHCP, GPON (OLT/ONT)

**Languages:** Spanish (Native) · English (B2)